BATCH 9 · GROUP SUBMISSION

Credit Pre-Screening Engine — Insight Summary

Calibo AI Academy · Path 2 · KLU Batch · Phase 2, Mini Use Case 02 · BFSI — Retail Lending

What we were asked. A mid-size private bank receives about 3,000 personal loan applications a month. Every one gets a manual review of roughly 25 minutes — about 1,250 analyst hours. Close to half are declined, and many of those are obvious within the first minute. We were asked whether six months of past decisions (12,000 applications) can safely remove the clear-cut declines from that queue, and at what cost in wrongly rejected customers.

Three key findings

1. The bank already has an unwritten credit score floor at 650. Below 650, only 3.0% of 1,179 applications were approved; immediately above it, 32.5%, rising to 85.2% at 800+. That is a tenfold jump across a single boundary — a rule rather than a gradient, and exactly the kind of rule a machine can apply with confidence.

2. Affordability is the strongest driver, and it breaks at 50% FOIR. Approval falls gently to 47.4% up to 50% FOIR, then collapses — 31.7% in the 50–55% band and 15.6% above 55%. FOIR is also the model's largest coefficient (+1.22). We propose writing a 50% FOIR limit, with 50–55% as a referral band, into credit policy; the bank's own past decisions already support it.

3. The model cannot read new-to-credit applicants. It is 81.2% accurate on applicants who have a credit score and 56.7% on those who do not — barely better than a coin toss. The cause is structural, not statistical: for all 1,039 new-to-credit applicants, delinquencies, active loans and existing EMIs are all zero, so three of the model's four strongest features carry no information about them.

Our recommendation

Auto-decline an application only when the model is at least **90% confident**. That removes **15.2% of applications — about 456 a month, roughly 190 analyst hours** — and 4.7% of those would have been approved by the credit team, just inside the bank's 5% risk limit. The default 0.5 cut-off would remove 44.9% but wrongly decline 20.9%, which no lending unit could sign off.

We also recommend **never auto-declining an applicant with no credit history**. Routing that group to a human lets the threshold drop to 0.88 and frees about **205 hours a month at a lower 4.1% wrong-decline rate** — more volume and less risk at the same time, and a rule the bank can explain to a regulator in one sentence.

One limitation

The target is the credit team's past decision, not whether the customer repaid. The model learns to copy historical analyst judgement, including any inconsistency or bias in it, so "4.7% wrong" means "4.7% disagreed with an analyst" — not "4.7% were good credit risks". Before going live the bank should shadow-run the lane for a month and start capturing repayment outcomes.

Who owned which section

Member	Role	Sections owned
Mohammed Mutee Ruknuddin	Data Analyst	Step 5.1 cleaning · CBIM Canvas
Mohammed Saalif Udyawar	Insights Analyst	Steps 5.2–5.3 — drivers, affordability
Mulla Javeed	Model Builder	Steps 5.4–5.5 — model, threshold
Nagothu Varshitha	Policy Synthesiser	Steps 5.6–5.7 — fairness, briefing